Prajwal Koirala

Nationality: Nepalese

(+977) 9845082214

Date of birth: 07/09/2000

Email address: 074bme627.prajwal@pcampus.edu.np

Website: https://prajwalkoirala.github.io/

Address: Ward - 11, Bharatpur Metropolitan City, Nepal, 44200 Bharatpur (Nepal)

EDUCATION AND TRAINING

BE in Mechanical Engineering

Pulchowk Campus, Institute of Engineering, Tribhuvan University [2017 – Current]

Address: Lalitpur (Nepal)

pcampus.edu.np

Calculus, Linear Algebra, Numerical Methods, Control Systems, Dynamics, Probability and Statistics, Instrumentation, Fluid Mechanics, Solid Mechanics, Electric Machines, Operations Research, Finite Elements Analysis

+2 Science (Grade XI and XII)

The Times International College [2015 – 2017]

Address: Dillibazar, Kathmandu (Nepal)

www.timescollege.edu.np

Final grade : 86.9%

Physics, Mathematics, Chemistry, Computer Science

School Leaving Certificate

Narayani Model Secondary School [2015]

Address: Bharatpur (Nepal)

Final grade : 88.5%

WORK EXPERIENCE

Research Intern

NSDevil [12/2021 – Current]

- www.nsdevil.com

Country: South Korea

(Remotely working from Nepal's office)

NSD-ROBO project

- ROS Development
- Autonomous Navigation
- Electronic circuitry

Assistant Course Instructor
Pi Academy [05/2021 – 10/2021]

- piacademy.edu.np

City: Kathmandu

Country: Nepal

(Taught classes on high school mathematics for engineering aspirants.)
Mathematics for Engineering Entrance Preparation

- Limit and Continuity of functions
- Derivatives and Antiderivatives

Industrial Attachment
Ebolt Technology [04/2021 – 05/2021]

- www.eboltmobility.com

City: Lalitpur

Country: Nepal

E-scooters

- Vibrations, Suspension System, MATLAB Simulations

Engineering Intern
NEAR Aerospace [09/2020 – 12/2020]

City: Lalitpur

Country: Nepal

Sounding Rocket

- Dynamics, Control and Estimation, MATLAB Simulations

DIGITAL SKILLS

Programming Languages
Programming Languages: C, Python, MATLAB, Javascript

CAD/CAM
Autodesk Fusion 360 / AutoCAD

Engineering Simulations
Ansys

Spreadsheet Modelling
Microsoft Excel

Hands-on experience with:
Arduino / Tensorflow / CasADi / Mathworks Simulink / Robot Operating System

LANGUAGE SKILLS

Mother tongue(s): **Nepali**

Other language(s): **English** | **Hindi**

AWARDS AND SCHOLARSHIPS

Scholarships

- Merit-based scholarship for BE studies at Pulchowk Campus, **Institute of Engineering** (2017)
- Merit-based scholarship for Grade XI and XII, **National Examination Board** (2015)

Awards (National Level Competitions)

- Winner, Model Based Design In MATLAB/Simulink, Hackathon by DELTA 2.0 (2021)
- Runner-up, Jigyasa: National Engineering Expo (2019)
- Highest Scorer, +2 Science Stream, National Examination Board, TTIC (2017)
- Winner, National Mathematics Quiz/Olympiad, At Budhanilkantha School (2016)

CERTIFICATIONS

Specializations

"Analytics for Decision Making" by University of Minnesota, Coursera, December 2021

- Introduction to Predictive Modeling
- Optimization for Decision Making
- Advanced Models for Decision Making
- Simulation Models for Decision Making

"Robotics" Specialization by University of Pennsylvania, Coursera, July 2020

- Aerial Robotics
- Computational Motion Planning
- Mobility
- Perception
- Estimation and Learning
- Capstone

"Deep Learning" Specialization by deeplearning.ai, Coursera, June 2020

- Neural Networks and Deep Learning
- Improving Deep Neural Networks: Hyperparameter Tuning, Regularization and Optimization
- Structuring Machine Learning Projects
- Convolutional Neural Networks
- Sequence Models

Courses

"Advanced AI Techniques for the Supply Chain" Course by LearnQuest, December 2021

"Control of Mobile Robots" Course by GeorgiaTech, December 2019

"Introduction to Mathematical Thinking" Course by Stanford University, October 2019

"Machine Learning" Course by Stanford University, December 2018

VOLUNTEERING

Mathematics Tutor

[Bharatpur, Chitwan, 2017 – 2019]

Tutored High School mathematics to the students of my hometown during vacations.

The pre-engineering course is designed to help students in university entrance examination which is competitive and prestigious in Nepal.

Volunteer

[Bharatpur, Chitwan, 2017 – 2018]

Actively participated as a volunteer in different sessions of "Shakriya - Public Speaking and Student Empowerment Program".

PROJECTS

Optimal Trajectory Generation for Autonomous Navigation of Wheeled Robots

[2021 – Current]

Final Year Capstone Project

- Supervisors: Prof. Dr. Mahesh Chandra Luitel and Asst. Prof. Dr. Sanjeev Maharjan
- Project incubated by *NSDevil, South Korea*
- Trajectory Generation formulated as a Non-Linear Optimization Problem
- Control of a wheeled robot using real-time information updates

prajwalkoirala.github.io/files/MidTermReport.pdf

Design and Analysis of an Axial Flow Compressor

[2021]
prajwalkoirala.github.io/files/AxialComp.pdf

Design and Control of Hydraulic Suspension System for Passenger Vehicles

[2020]
prajwalkoirala.github.io/files/HydraulicSuspension.pdf

Design of Machine Elements for a Two-Stage Plastic Shredder

[2020]
prajwalkoirala.github.io/files/Two-Stage-Plastic-Shredder-MD-Report.pdf

Self driving robotic car using Artificial Neural Network

[2019]
github.com/PrajwalKoirala/Self-driving-robotic-car-using-Artificial-Neural-Network

Other Projects

Other projects are listed on my website.

prajwalkoirala.github.io/